

Practical Simulation Test: Forest Firefighting

Robotics Mission

Course: Computer Science, Undergraduate Robotics

Platforms: Webots Simulator + Forest Firefighters Project

GitHub Reference:

https://github.com/cyberbotics/webots-projects/tree/master/projects/forest_firefighters

Objective

Students will program one or more firefighter robots (drone or legged robot) to autonomously detect, navigate toward, and extinguish forest fires in a dynamic environment. The task integrates locomotion, perception & sensing, and navigation.

Available Robots

- Mavic 2 Pro drone (aerial locomotion + camera)
- Spot-like legged robot (ground locomotion + camera + water dropper)

Mission Requirements

Your robot(s) must:

1. Patrol a designated forest area.
2. Detect fire/smoke using a camera (perception & sensing).
3. Navigate efficiently to the fire location.
4. Extinguish the fire by dropping water accurately.
5. Handle multiple fires if they appear (propagation).

Constraints

- Simulation time limit: 5 - 8 minutes (adjustable).
- Robots start from a base position.
- Wind and fire propagation are enabled (realistic challenges).
- You may use one robot or a team (drone + ground robot).

Required Implementations

Component	Key Features Students Must Implement	Weight
Locomotion	- Stable movement (drone: altitude control, velocity; Legged: gait/walking) - Obstacle avoidance during movement - Precise positioning for water drop	25%
Perception & Sensing	- Camera image processing (OpenCV recommended) to detect fire/smoke by colour - Optional: Distance estimation to the fire - Sensor fusion if multiple sensors are used (e.g GPS + camera)	30%
Navigation	- Path planning to the fire location. - Localisation (GPS/odometry) - Dynamic re-planning if fire spreads or new fires appear.	25%
Integration & Behavior	- Overall autonomous control loop - Decision making (patrol → detect → go → extinguish → return) - Handling of multiple fires	10%
Code Quality & Documentation	- Clean, well-commented code - Report (2–4 pages) explaining methods, challenges, and results	10%

Total		100 marks
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Deliverables

1. Controller Code (Python) placed in the appropriate controllers/ folder.
2. Report (PDF) including:
 - System architecture diagram
 - Key algorithms used
 - Challenges faced & solutions
 - Performance results (fires extinguished, time taken)
3. Presentation: live demo + explanation.

Guiding Steps

- A. Set up Webots + understand the Forest Firefighters project
- B. Implement perception (fire detection)
- C. Locomotion + basic navigation
- D. Full integration + testing
- E. Submission

“All the best”